Amylase and pH: Continuous-Sampling Investigation

I can follow the centre-approved Pearson method to investigate how pH affects amylase activity, recording safe continuous-sampling evidence accurately.

IODINE WELLS
prepared first

FIXED INTERVAL
continuous sample

RAW RESULTS
preserve evidence

Open the original interactive lesson

What success looks like

- Follow the authorised method and role boundaries, including iodine and temperature-control precautions.
- Sample at the agreed interval and recognise iodine results showing starch detected and starch not detected in the sampled drop.
- Record pH, endpoint time and observations without changing or hiding an uncertain result.

Retrieve and reconnect

- State the independent and dependent variables.
- Name two variables that must stay constant.
- State what the iodine endpoint means.
- Lead-in: explain why all iodine wells should be prepared before the reaction starts.

A group starts the reaction and only then begins adding iodine to the spotting tile. What is your first judgement?

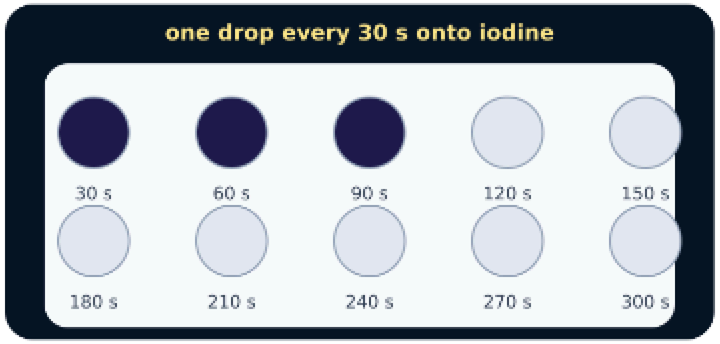
Think first. Point, say, sign or write your choice. Give one reason.

A. The method is fine because the reaction can wait while the tile is prepared.

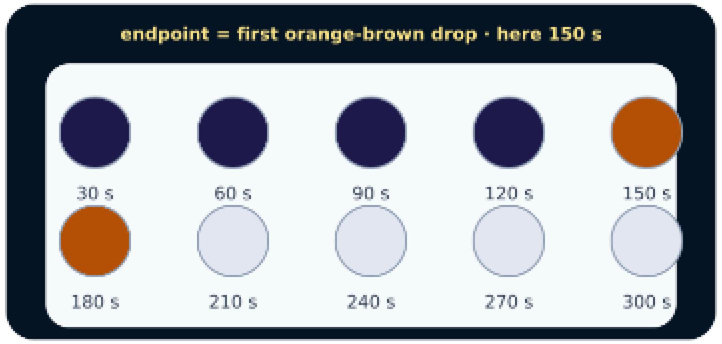
B. Pause and reset: prepare the iodine wells first so sampling can begin at the agreed time.

C. Add extra amylase to make up for the delay.

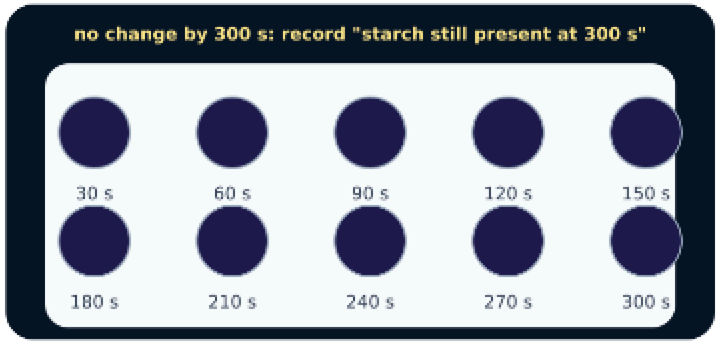
Look closely · what is the evidence?



Spotting tile with three blue-black drops



Tile showing the change from blue-black to orange-brown at 150 s



Tile with all ten drops blue-black

Words we will use

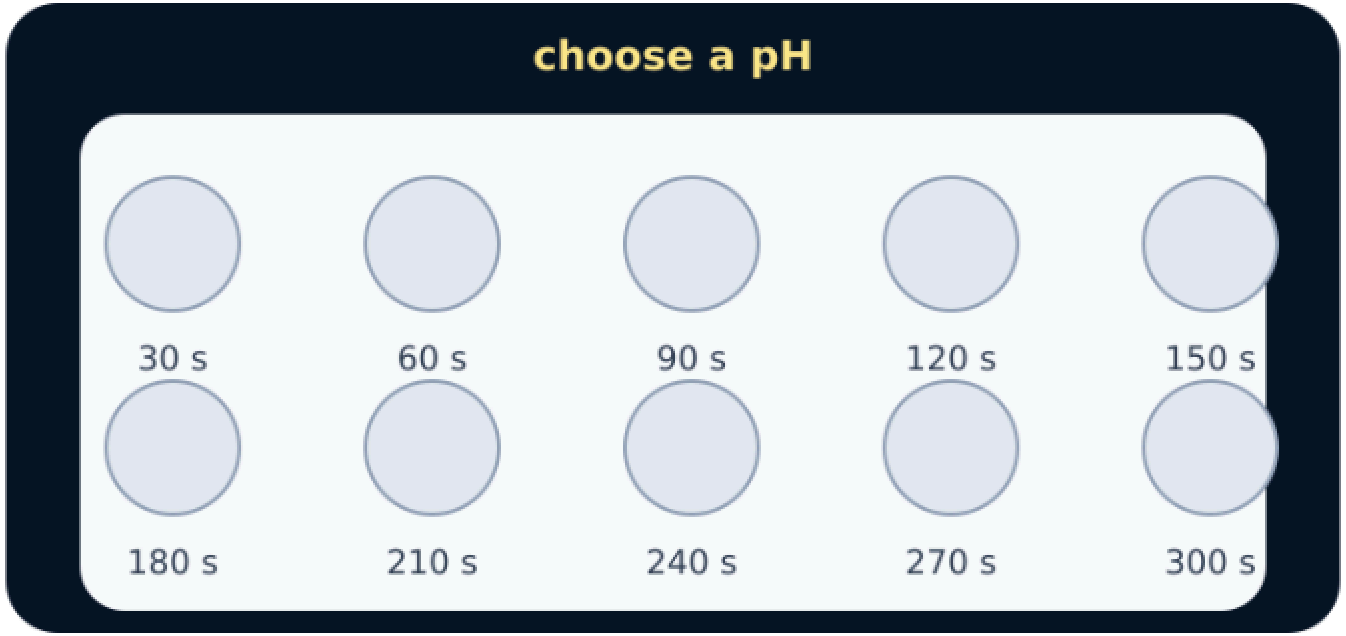
- continuous sampling — test small samples over time
- endpoint — the stopping observation
- validity — tests the intended question

Watch the model

- Teacher checks goggles, labelled solutions, tile layout, clean transfer equipment, timing roles and the authorised temperature-control arrangement before permission to begin.

- Prepare the specified small iodine volume in each labelled tile well before mixing amylase, buffer and starch in the authorised order.
- Start timing at the exact method-defined event. At each fixed interval transfer one clean sample to a new iodine well and record the colour honestly.

Watch the model



Spotting tile filling with iodine drops

At 60 s the iodine well is blue-black; at 90 s it remains orange-brown. What endpoint time should be recorded with this sampling interval?

A. 60 s, because starch was definitely present then.

B. 90 s, the first sampled time at which starch was not detected, with interval uncertainty acknowledged.

C. 75 s exactly, because the colour must have changed at the midpoint.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Keep roles clear: one trained pupil samples, one calls the interval, one records, and one checks the tile map; rotate roles only when the teacher says it is safe.

- Use separate labelled transfer equipment as directed to reduce cross-contamination. Never return a sample to a stock solution or reaction tube.

- If a sample is missed, a colour is ambiguous or equipment is contaminated, preserve the raw record, tell the adult and mark the result for repeat or cautious interpretation rather than fabricating a value.

Find and repair the misconception

Evidence glitch: if the colour is hard to judge, record the time that makes the graph look smooth.

A. Keep it: a smooth graph is more scientific.

B. Repair it: record the observed uncertainty and repeat if authorised; never choose a preferred value.

C. Repair it: delete the entire table so no one sees the uncertain trial.

Method-decision checkpoint

Before or during the physical practical, classify each situation as continue, stop/check, or record-and-repeat.

- Choose one live-lab situation.
- Commit it to the safest scientific response.
- Use feedback to repair role or evidence decisions.
- Freeze the checkpoint, then carry out only the authorised physical step.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- CONTINUE AS AUTHORISED
- STOP AND CHECK ADULT
- RECORD, THEN REPEAT IF AUTHORISED

Name the condition that allowed continuation or forced a stop/repeat, and link it to safety, validity or honest evidence.

Your independent mission

Carry out the assigned part of the centre-approved amylase-pH investigation and preserve a complete raw evidence record.

Record: A dated raw table with pH, each sampled colour, first observed endpoint time, anomalies/uncertainty and the pupil's own method note. The page itself makes no CP-completion claim.

Choose your support and challenge

- Supported: Complete one clearly labelled reaction timeline with an adult-authorised role; circle the first orange-brown endpoint and state whether starch is detected.
- Standard: Run or record assigned pH trials, identify endpoint times and annotate one control-variable or uncertainty check.
- Stretch: Preserve full sampling evidence, estimate the maximum endpoint-time uncertainty from the interval and justify whether a repeat is needed.

Show what you know

State the endpoint for one trial and explain why a shorter time indicates greater amylase activity.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.